

# The perspectives of southern magnetic peculiar stars study with SALT telescope

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# Ap/Bp Si-stars

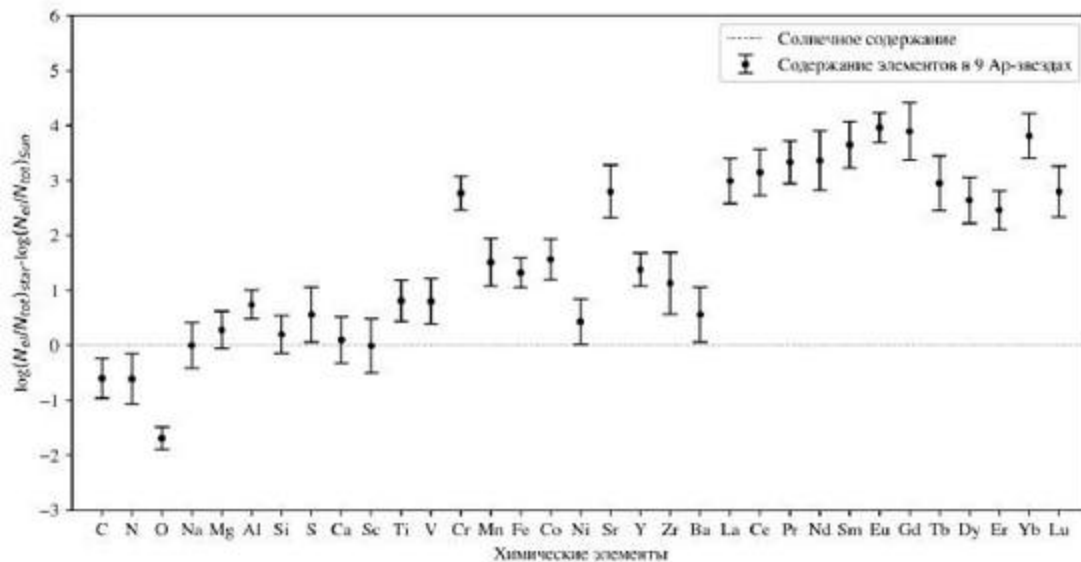
1

- Magnetic fields ~ 30 kG
- Abundance anomalies by 5 dex for heavy elements
- Vertical (stratification) and horizontal (spots) gradients of elements (theory: Michaud 1970, LeBlanc+2009, Stift&Alecian 2016, observations: Ryabchikova+2003)

Таблица 2: Классификация CP-звезд

| Интервал температур | Магнитные звезды         | Немагнитные звезды         |
|---------------------|--------------------------|----------------------------|
| 7000–10000          | Ap, SrCrEu<br>A3-F0      | Am, $\lambda$ Boo<br>A0-A1 |
| 10000–14000         | Ap Si<br>B8-A2           | Ap Hg-Mn<br>B6-B9          |
| 13000–18000         | He-weak Si,SrTi<br>B3-B7 | He-weak P Ga<br>B4-B5      |
| 18000–22000         | He-strong<br>B1-B2       |                            |

Romanuyk 2007



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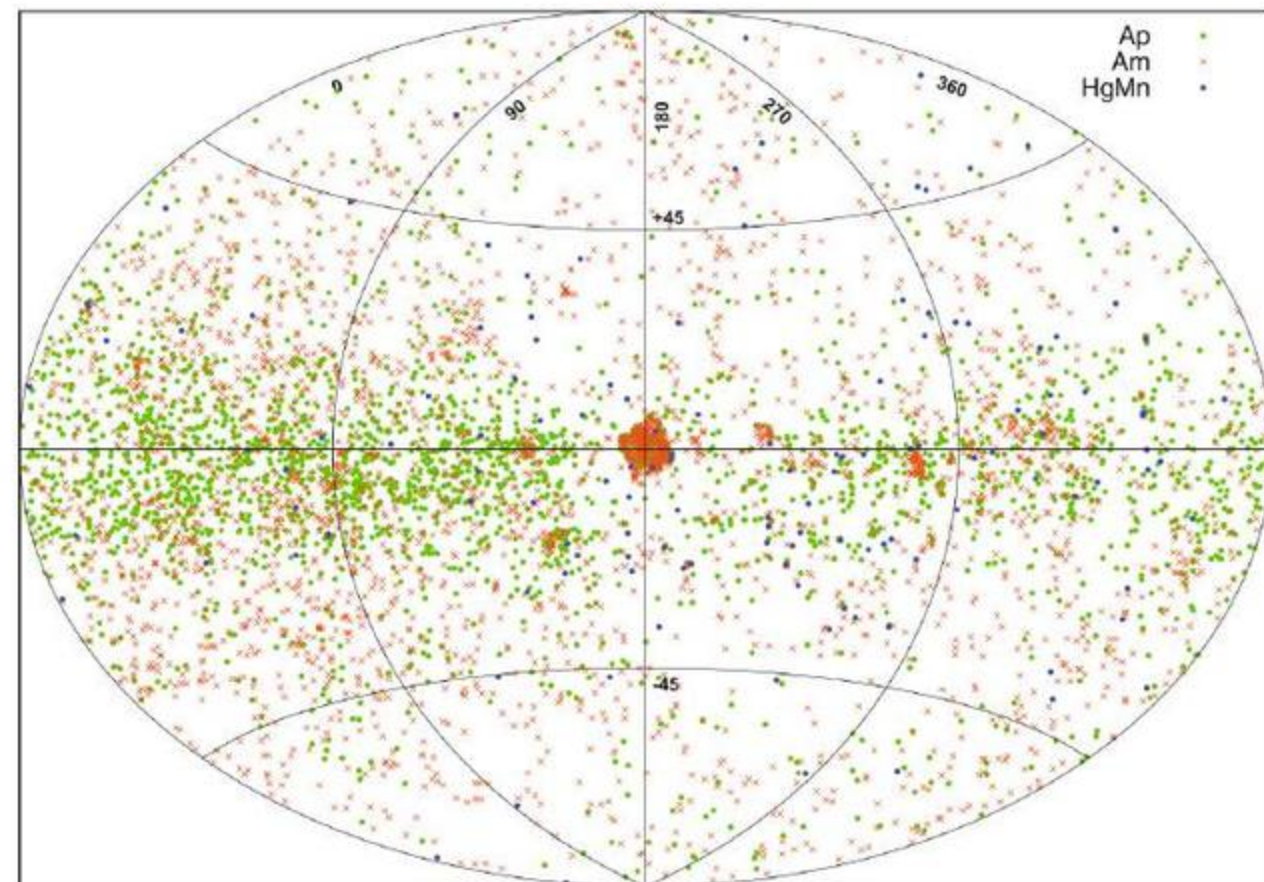
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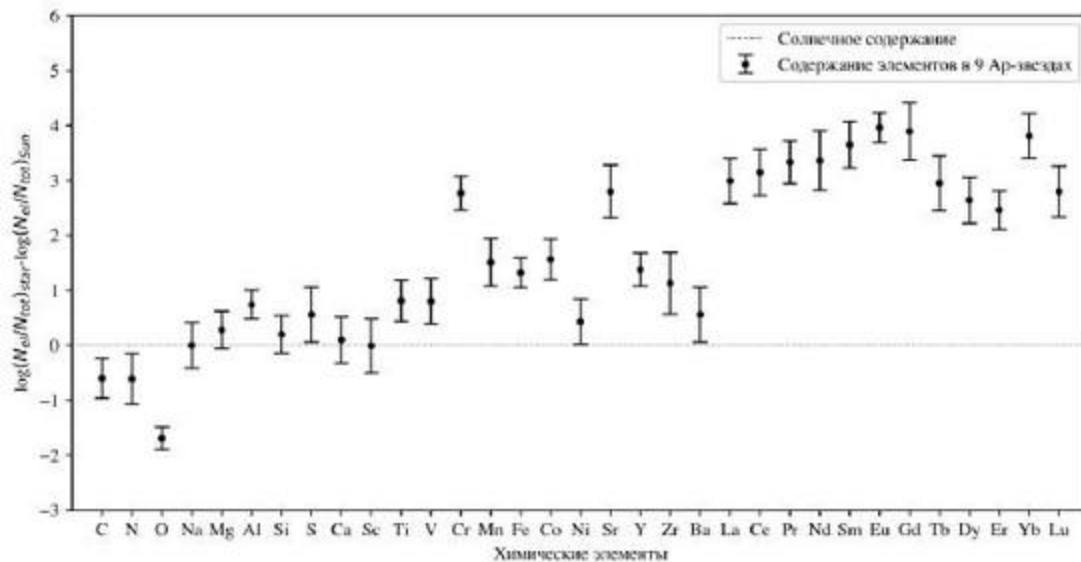
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Romanuyk 2007



Renson+2009





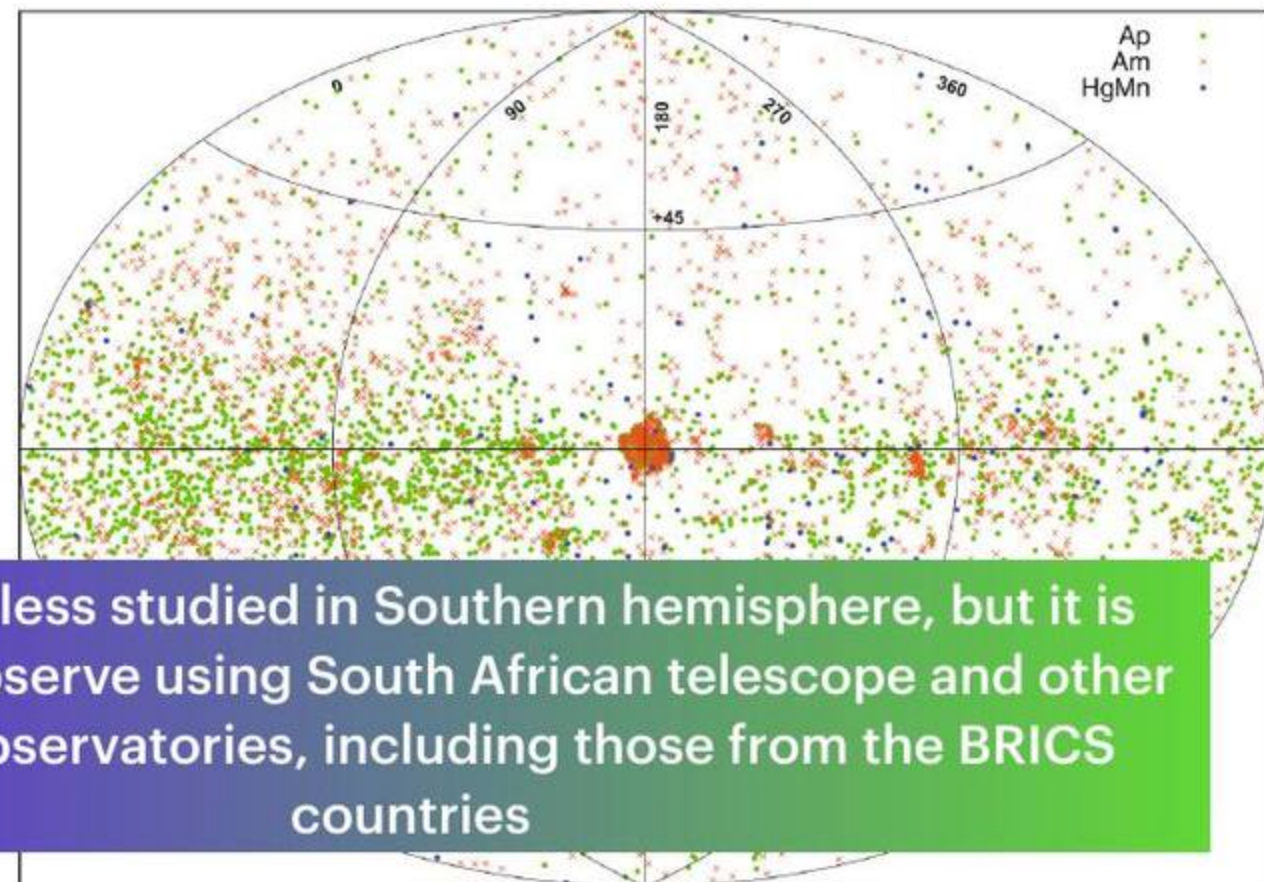
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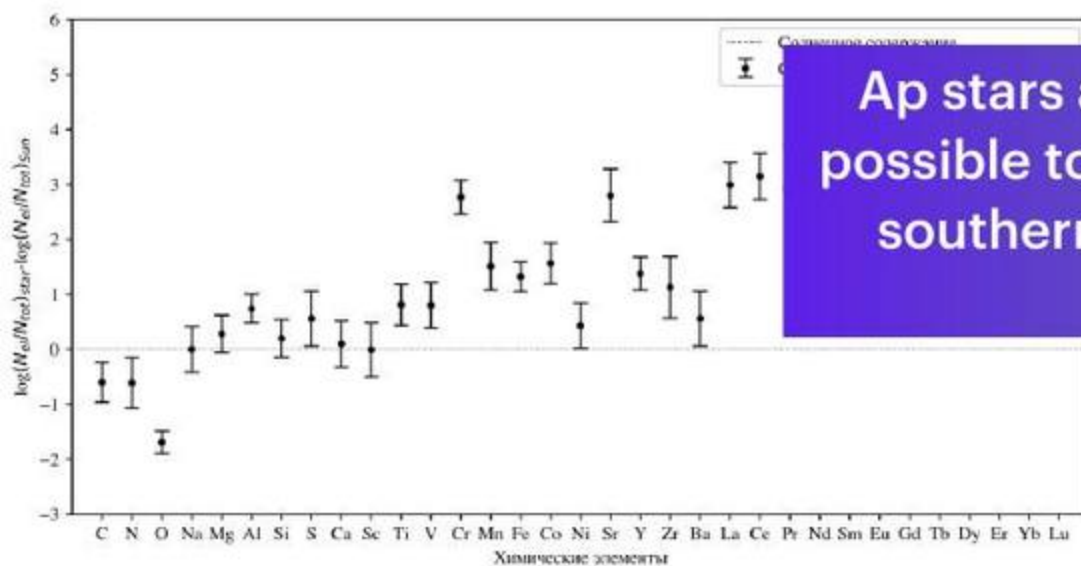
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Romanuyk 2007



Renson+2009





# Ap/Bp Si-stars

## V\* MX TrA -- alpha2 CVn Variable

Other object types: [?](#) \* (HD, CD, ...), \*\* (\*\*, WDS), a2\* (2003AstL), V\* (V\*), UV (TD1), NIR (2MASS), MIR (WISE)

ICRS coord. (*ep*=J2000) : 16 59 33.9601430494 -69 16 05.373376901 (Optical) [ 0.1242 0.1492 90 ] A 2020yCat.1350....0G

FK4 coord. (*ep*=B1950 *eq*=1950) : 16 54 10.7424252154 -69 11 31.968394539 [ 0.1242 0.1492 90 ]

Gal coord. (*ep*=J2000) : 321.9308858592135 -16.1829724177089 [ 0.1242 0.1492 90 ]

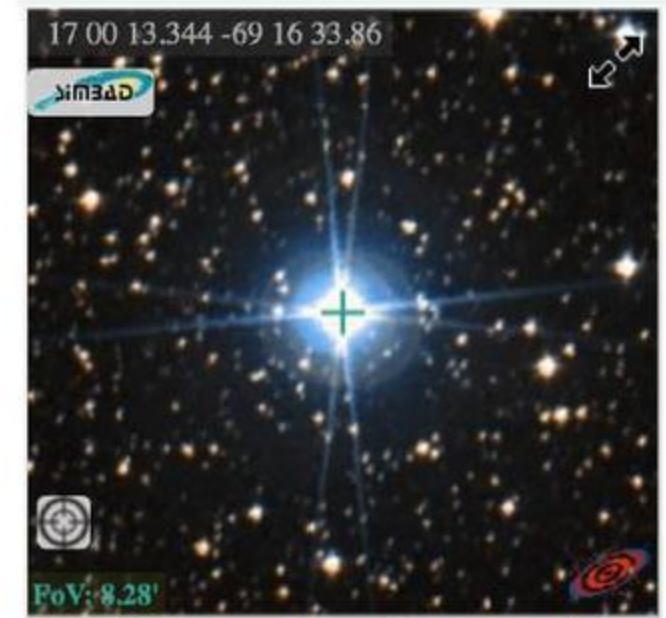
Proper motions *mas/yr* : -14.150 -13.233 [0.165 0.209 90] A 2020yCat.1350....0G

Radial velocity / Redshift / cz : V(km/s) -1.70 [2.1] / *z*(-) -0.000006 [0.000007] / cz -1.70 [2.10]  
C 2006AstL...32..759G

Parallaxes (*mas*): 5.2209 [0.2382] A 2020yCat.1350....0G

Spectral type: ApSi D 1975MSS...C01....0H

Fluxes (6) :  
B 5.77 [0.01] D 2002A&A...384..180F  
V 5.89 [0.01] D 2002A&A...384..180F  
G 5.871282 [0.002903] C 2022yCat.1355....0G  
J 5.902 [0.019] C 2003yCat.2246....0C  
H 5.943 [0.026] C 2003yCat.2246....0C  
K 5.943 [0.024] C 2003yCat.2246....0C



## Previous studies:

$$P_{rot} = 2.163^d \text{ (Paunzen \& Maitzen 1998)}$$

$$T_{eff} = 12500, \lg g = 3.76 \text{ (Netopil et al. 2017)}$$

$$v_e \sin i = 75 - 103 \text{ km/s (Levato et al. 1996, Netopil et al. 2017)}$$

The star was not studied using high-resolution spectroscopy and spectropolarimetry

# Ap/Bp Si-stars

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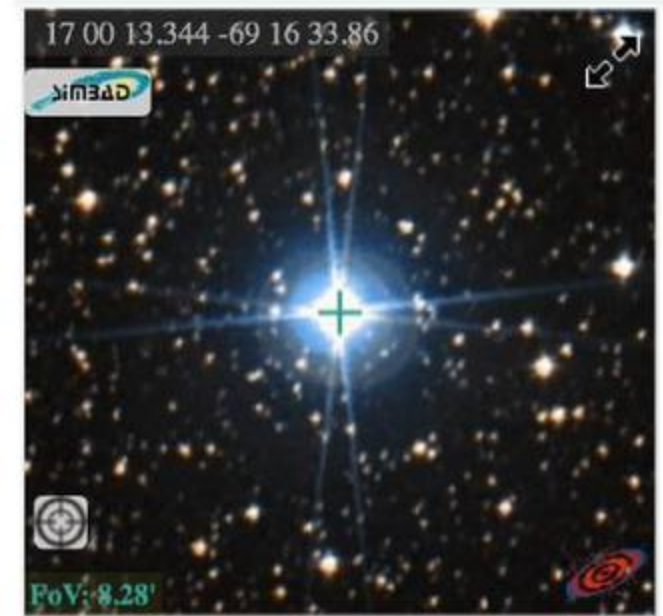
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## Observations from 2019 show:

- the star shows moderate rotational broadening
- significant line profile variability, modulated with the rotational phase
- a clear evidence for vertical abundance gradients
- A high rotation velocity allows for Doppler imaging of the surface



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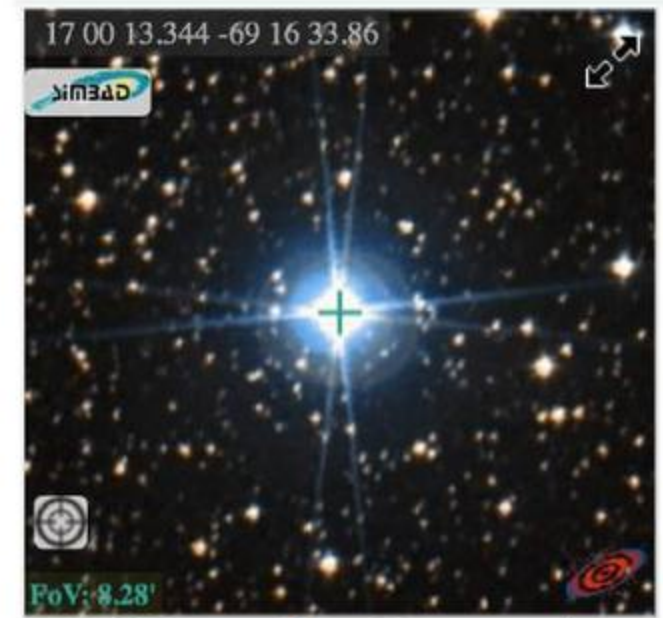
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## Aim of the study

It is a promising target for investigating the effects of inhomogeneous three-dimensional abundance distribution on the emergent radiation properties

## Steps of study

- Analysis of spectra of twin stars with different  $v_e \sin i$  MX TrA and BD+00° 1659, determination of the atmospheric parameters and abundance using LTE and non-LTE methods
- Perform the multi-element Doppler imaging of the surface of MX TrA
- Analysis of stratification of Fe and Si in BD+00 1659 atmosphere
- **Calculation synthetic light curve of MX TrA using DI maps and stratification profile**

## Observations

**MX TrA** spectrum — HRS (SALT):  
HR mode  $R = 67000\text{--}74000$ ,  
 $3735\text{--}8870 \text{ \AA}$

**BD +00°1659** spectrum —  
ESPaDOnS (CHFT):  $R = 68000$ ,  
 $3700\text{--}10500 \text{ \AA}$

Photometry — UV, optical, IR:

- TD1 (Thompson et al. 1978) - MX TrA
- GAIA EDR3 (Gaia Collaboration 2021), APASS9 (Henden+2016) - both
- 2MASS (Skrutskie+2006), WISE (Wright+2010) - both

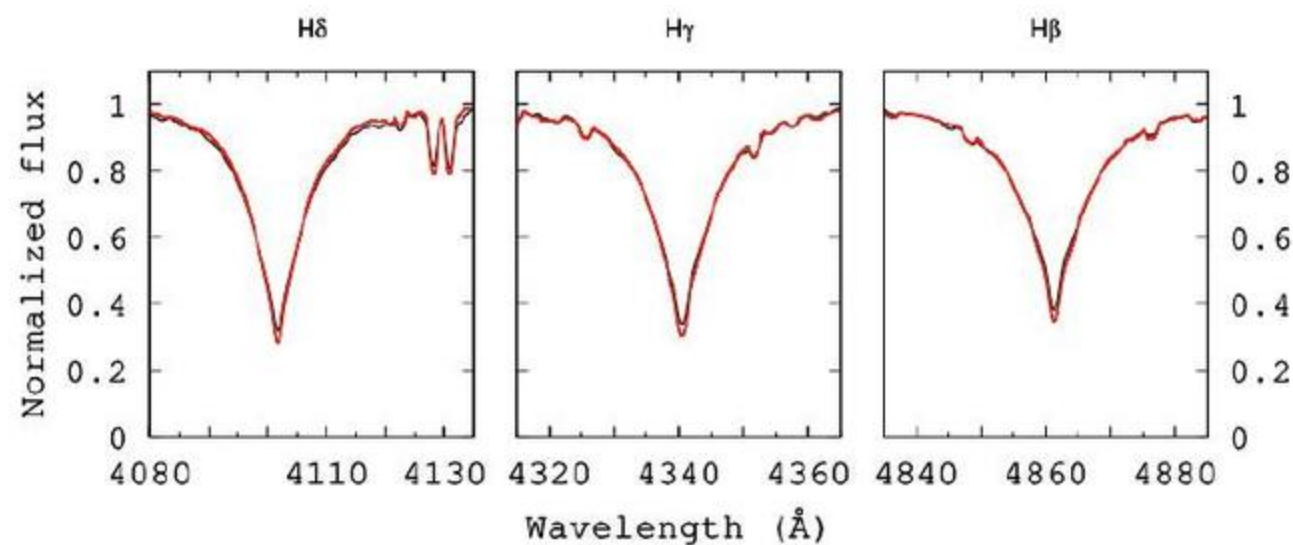
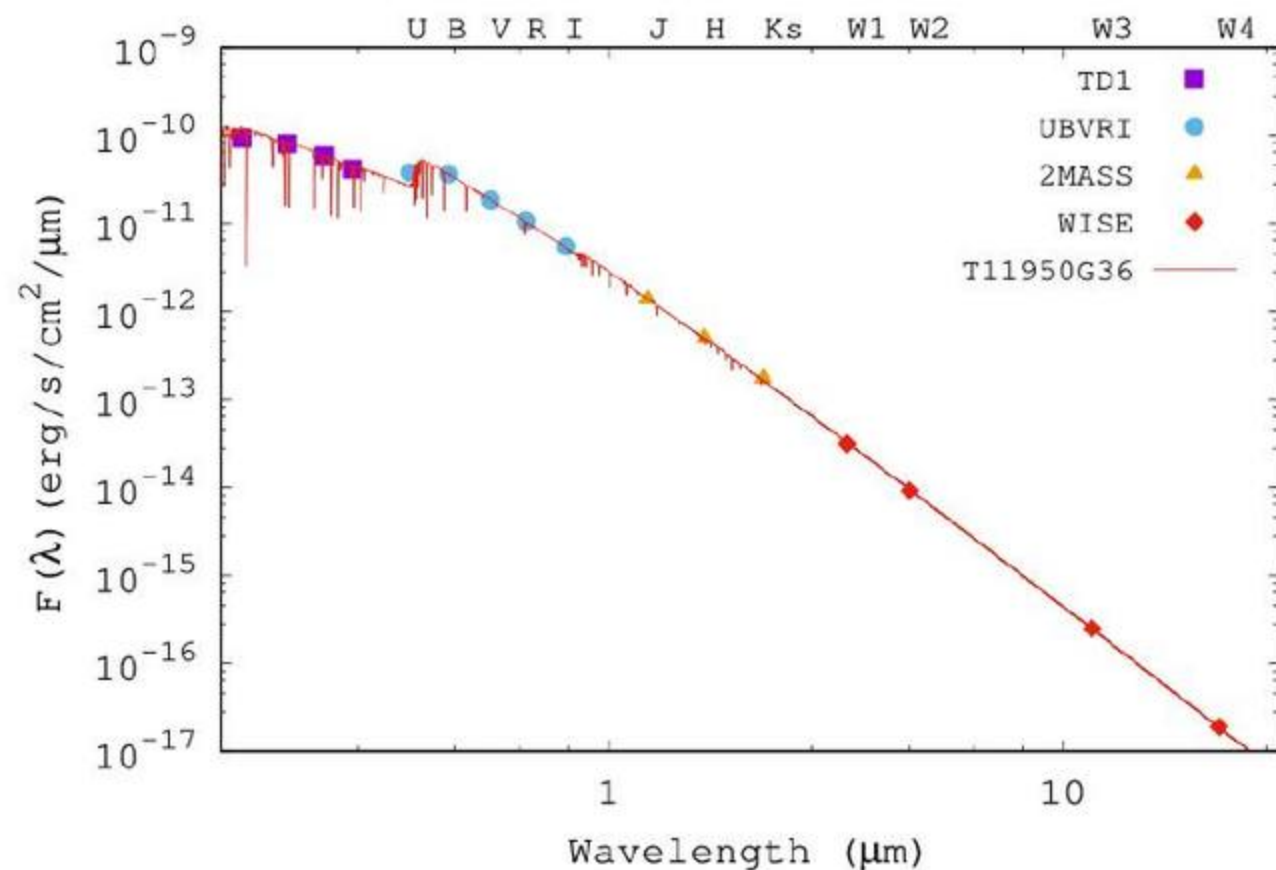


|                      |                                |
|----------------------|--------------------------------|
| $T_{\text{eff}}$     | $11\,950 \pm 200\text{ K}$     |
| $\log g$             | $3.6 \pm 0.2\text{ dex}$       |
| $\xi_t$              | $0.0\text{ km s}^{-1}$         |
| $\zeta_{\text{RT}}$  | $0.0 \pm 1.5\text{ km s}^{-1}$ |
| $v \sin i$           | $69 \pm 2\text{ km s}^{-1}$    |
| $\log (L/L_{\odot})$ | 2.42                           |
| $M/M_{\odot}$        | 2.1                            |
| $R/R_{\odot}$        | 3.8                            |

$$B < 2kG$$



From fitting of the Balmer line profiles and spectral energy distribution (SED)





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Table 3. Average LTE abundances in HD 152564

| Ion    | $N_{\text{lines}}$ | $\log(A)_X$        | $\log(A)_{\odot}^{\odot}_X$ |
|--------|--------------------|--------------------|-----------------------------|
| He I   | 6                  | $-1.61 \pm 0.09$   | -1.13                       |
| C II   | 3                  | $-2.70 \pm 0.04$   | -3.58                       |
| O I    | 2                  | $-4.03 \pm 0.14^*$ | -3.35                       |
| Mg II  | 4                  | $-4.97 \pm 0.08$   | -4.49                       |
| Al II  | 2                  | $-6.72 \pm 0.20$   | -5.61                       |
| Si II  | 11                 | $-3.62 \pm 0.22$   | -4.53                       |
| Si III | 3                  | $-2.55 \pm 0.13$   | -4.53                       |
| Ti II  | 6                  | $-6.13 \pm 0.14$   | -7.07                       |
| Cr II  | 7                  | $-5.50 \pm 0.16$   | -6.42                       |
| Fe I   | 4                  | $-4.50 \pm 0.25$   | -4.58                       |
| Fe II  | 12                 | $-3.85 \pm 0.12$   | -4.58                       |
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Note. \* - NLTE abundance is given for oxygen.

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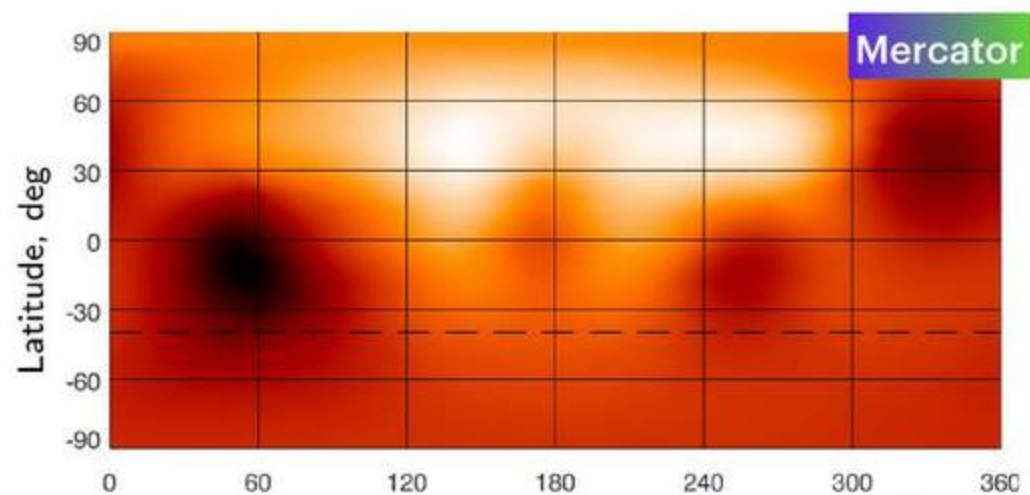
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Si II—III and Fe I—III anomaly - presence of stratification, but due to the  $v \sin i = 69$  km/s stratification analysis cannot be performed

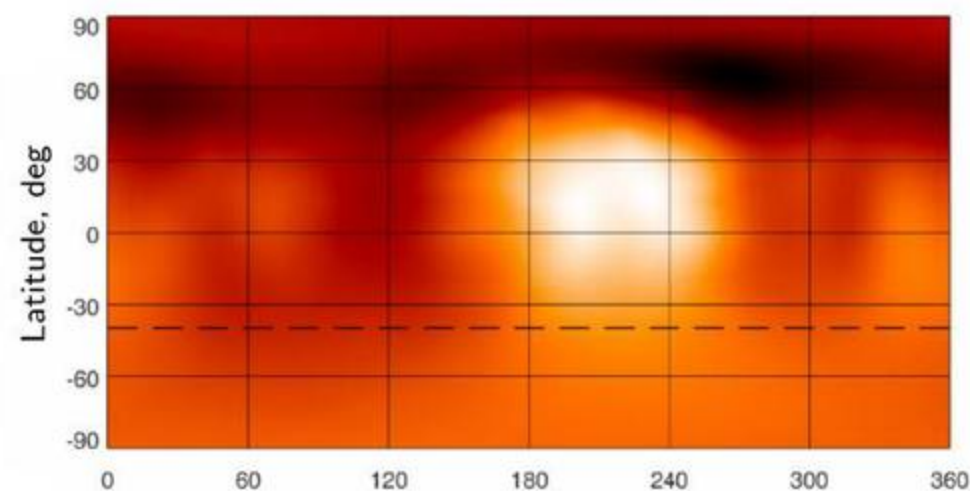
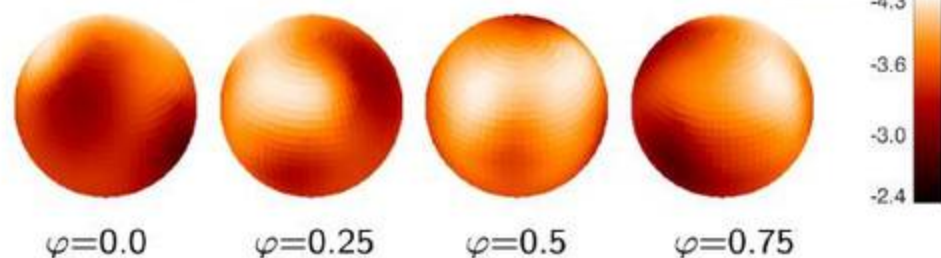




Si II 5041Å

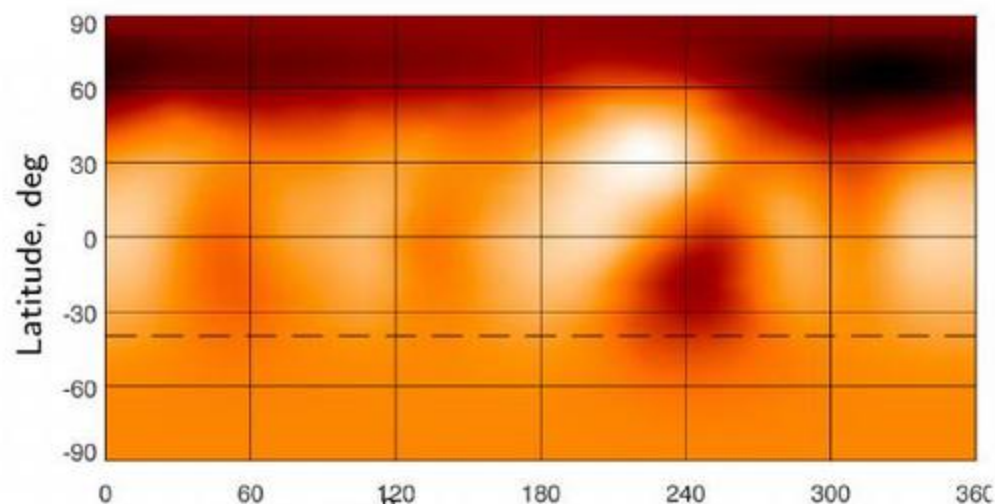
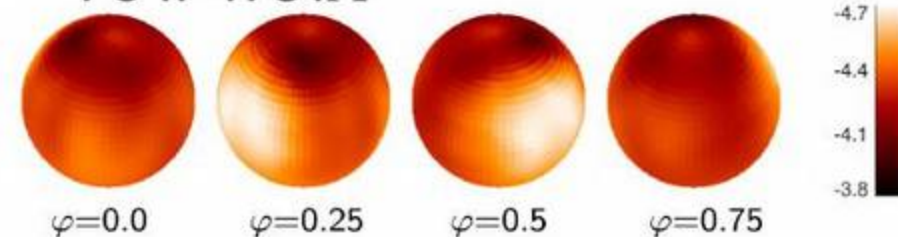
Longitude, deg

Spherical



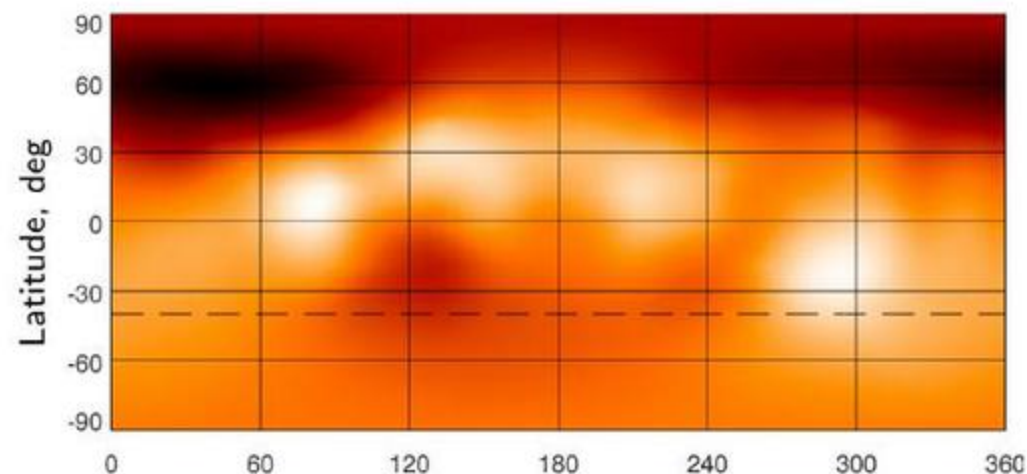
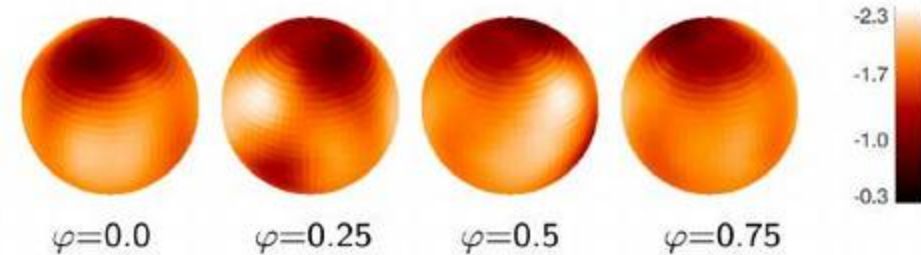
Fe II 4731Å

Longitude, deg



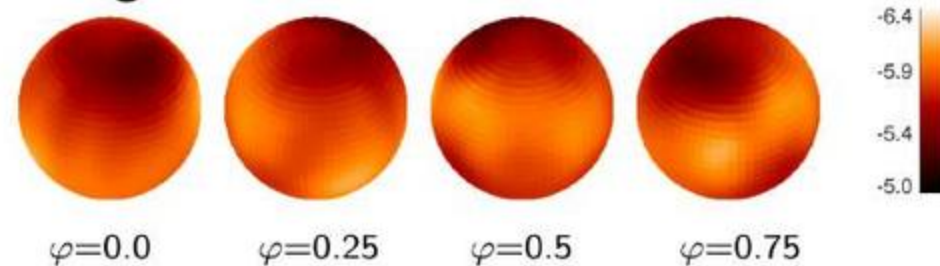
He II 5876Å

Longitude, deg

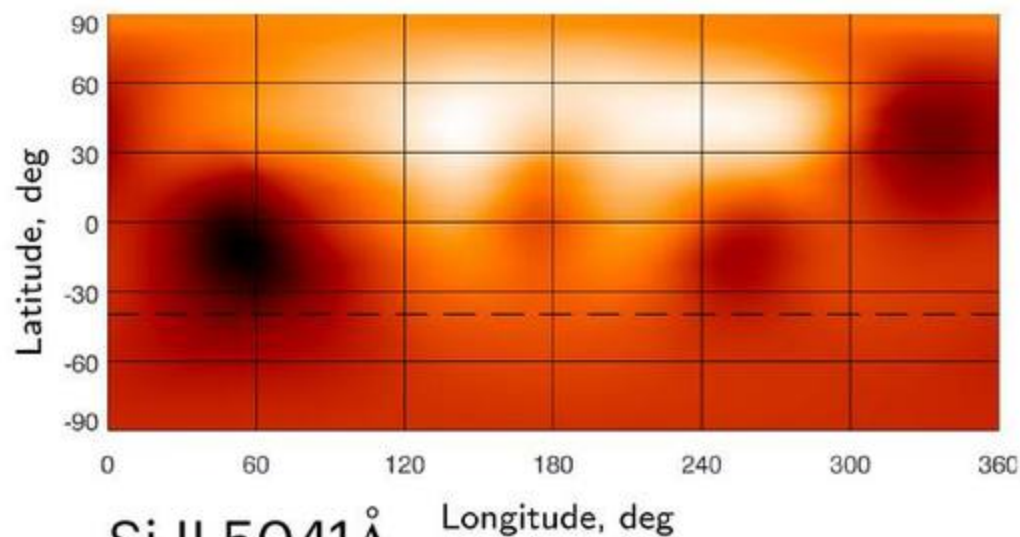


Mg II 4481Å

Longitude, deg

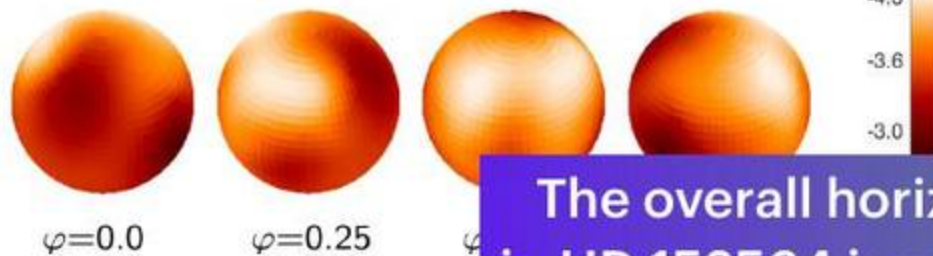
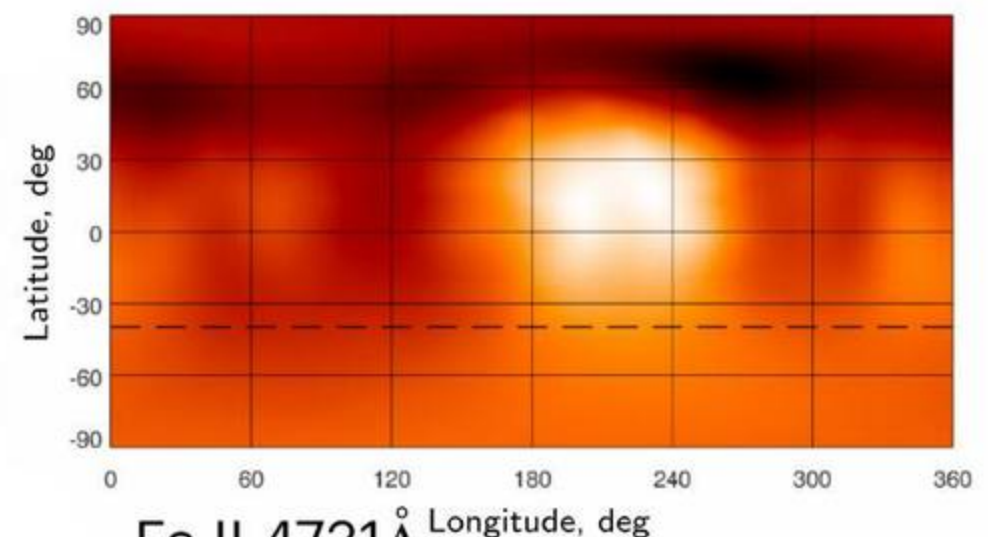






Si II 5041Å

Longitude, deg

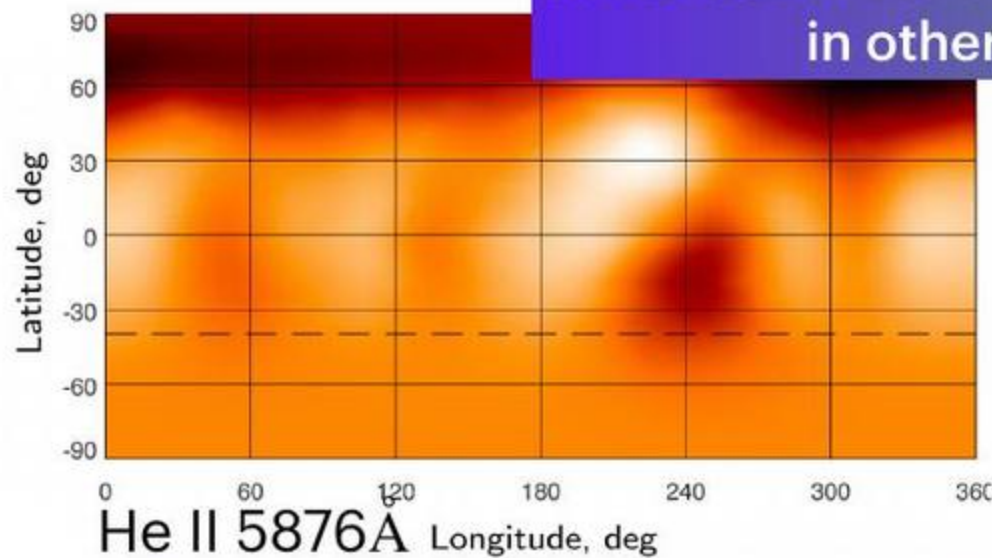
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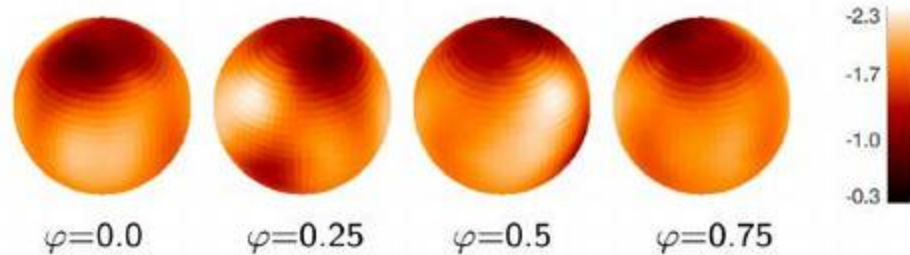
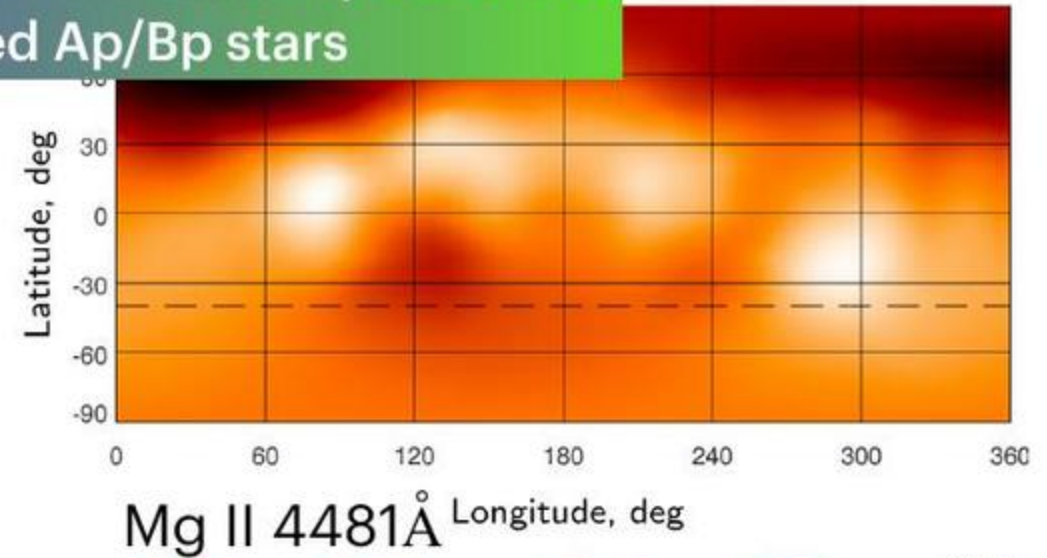
 $\varphi=0.75$ 

The overall horizontal distribution of elements in HD 152564 is well within the diversity observed in other mapped Ap/Bp stars



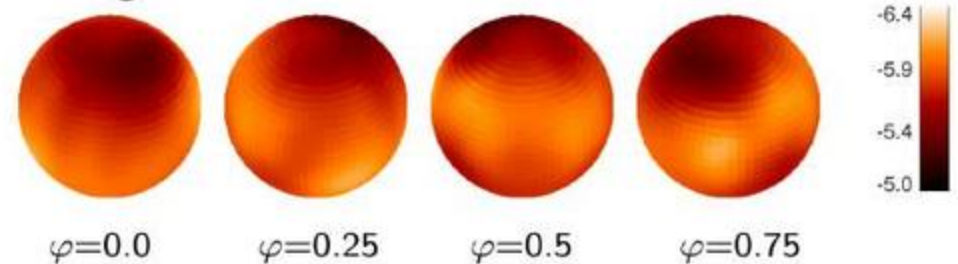
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 $\varphi=0.0$  $\varphi=0.25$  $\varphi=0.5$  $\varphi=0.75$ 

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Longitude, deg

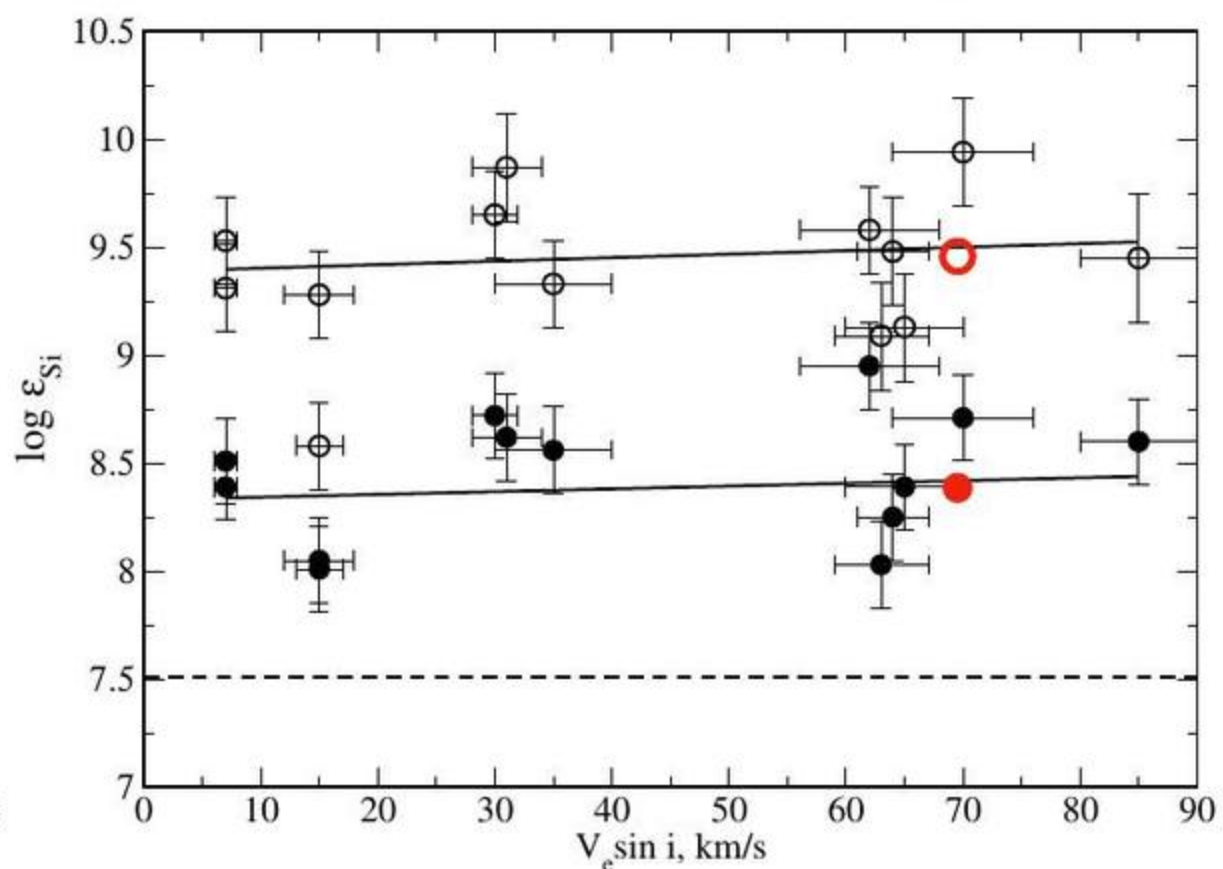
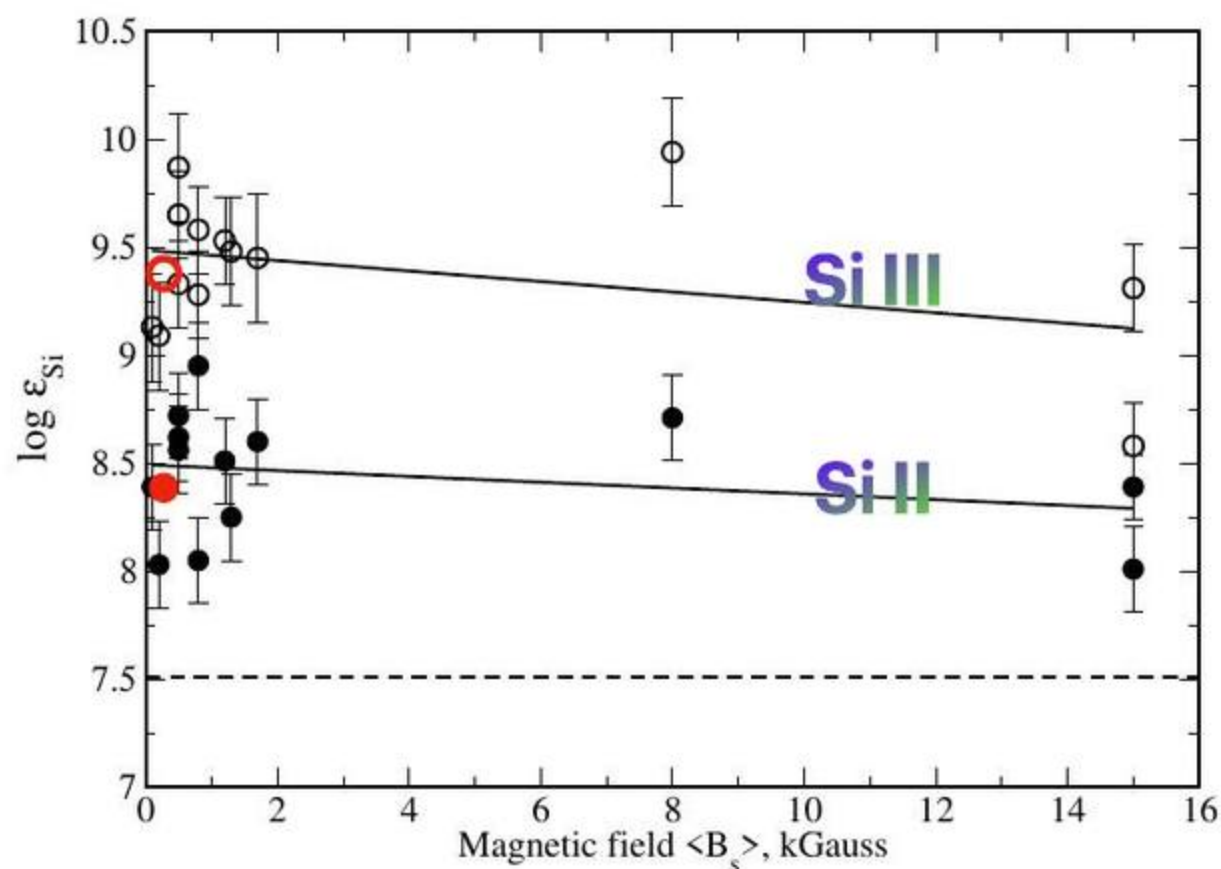
 $\varphi=0.0$  $\varphi=0.25$  $\varphi=0.5$  $\varphi=0.75$



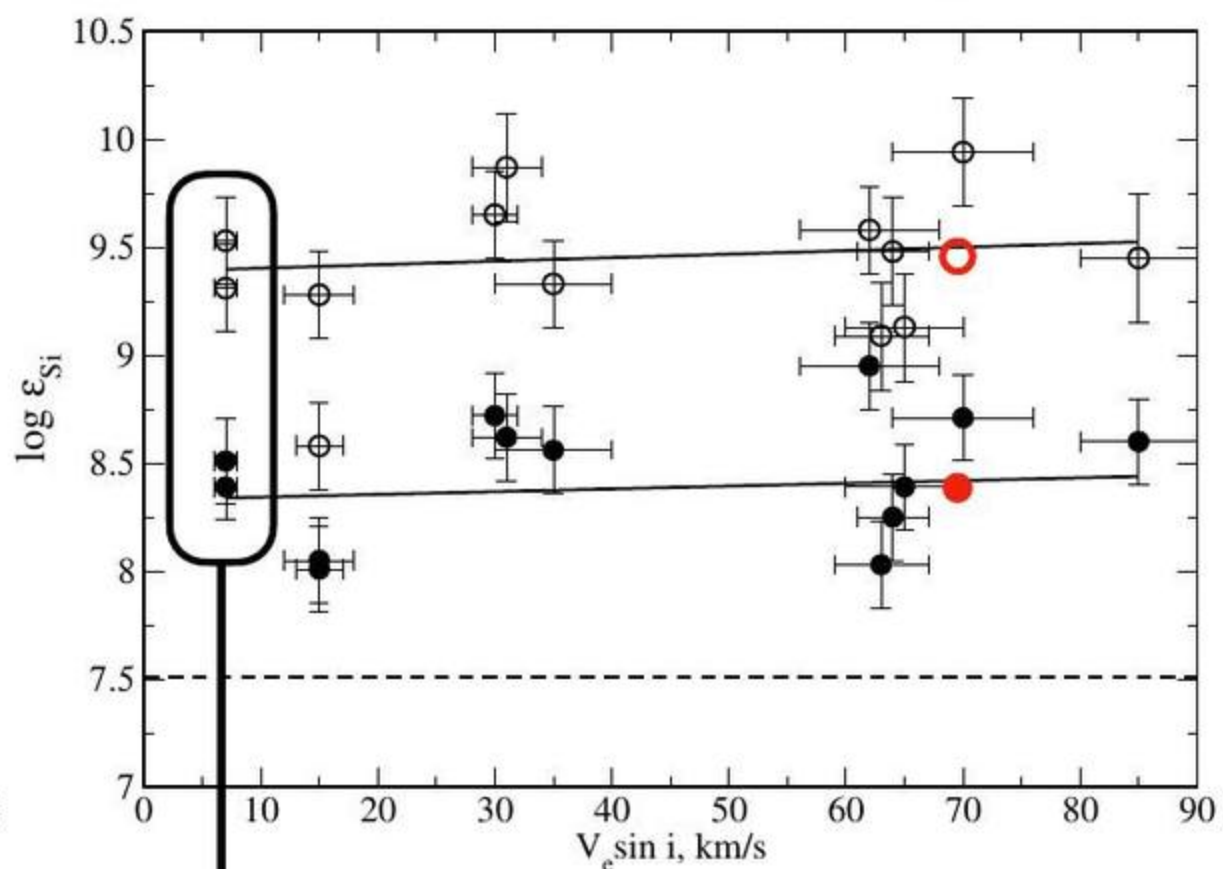
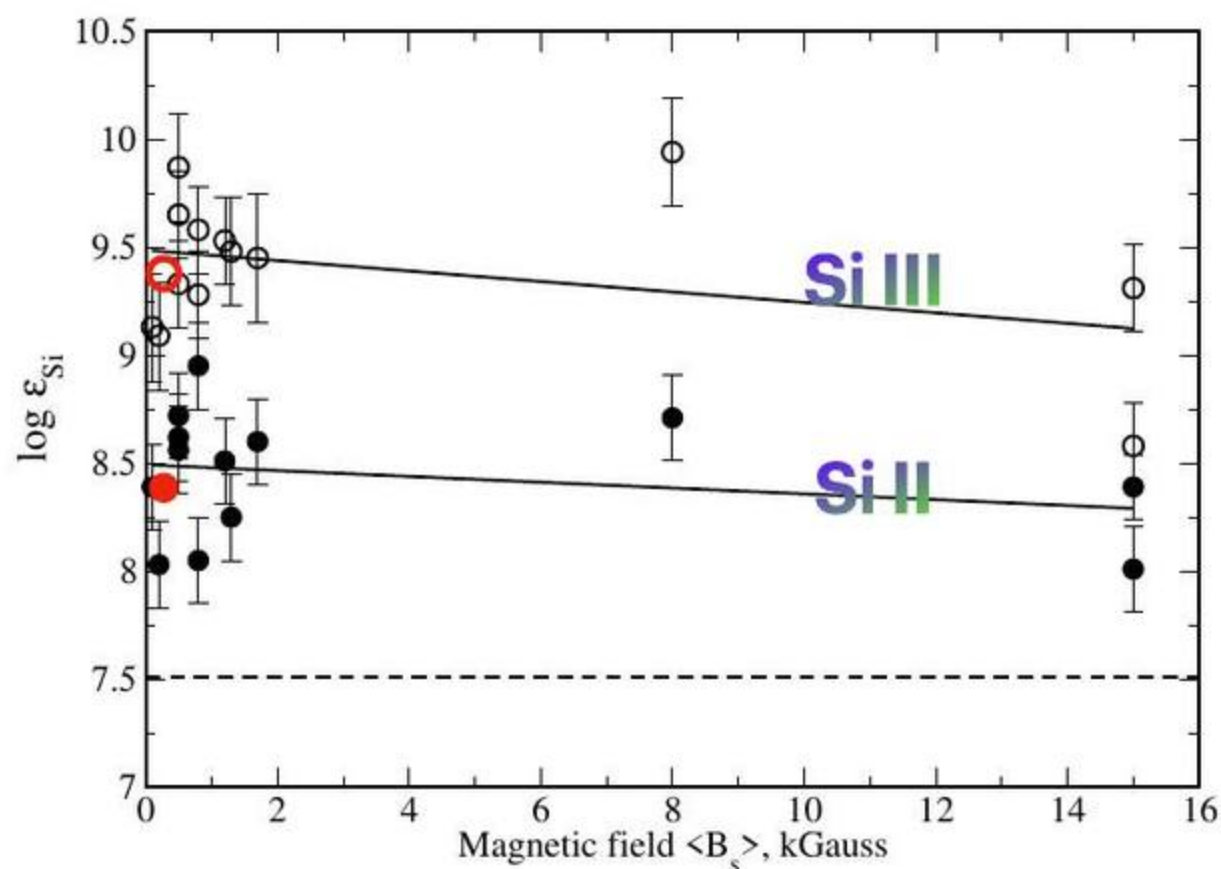
investigating the effects of inhomogeneous three-dimensional abundance distribution

From the stellar sample at Bailey&Landstreet 2013 - there is no dependence on  $\langle B_s \rangle$  and  $V_e \sin i$ :

MX TrA



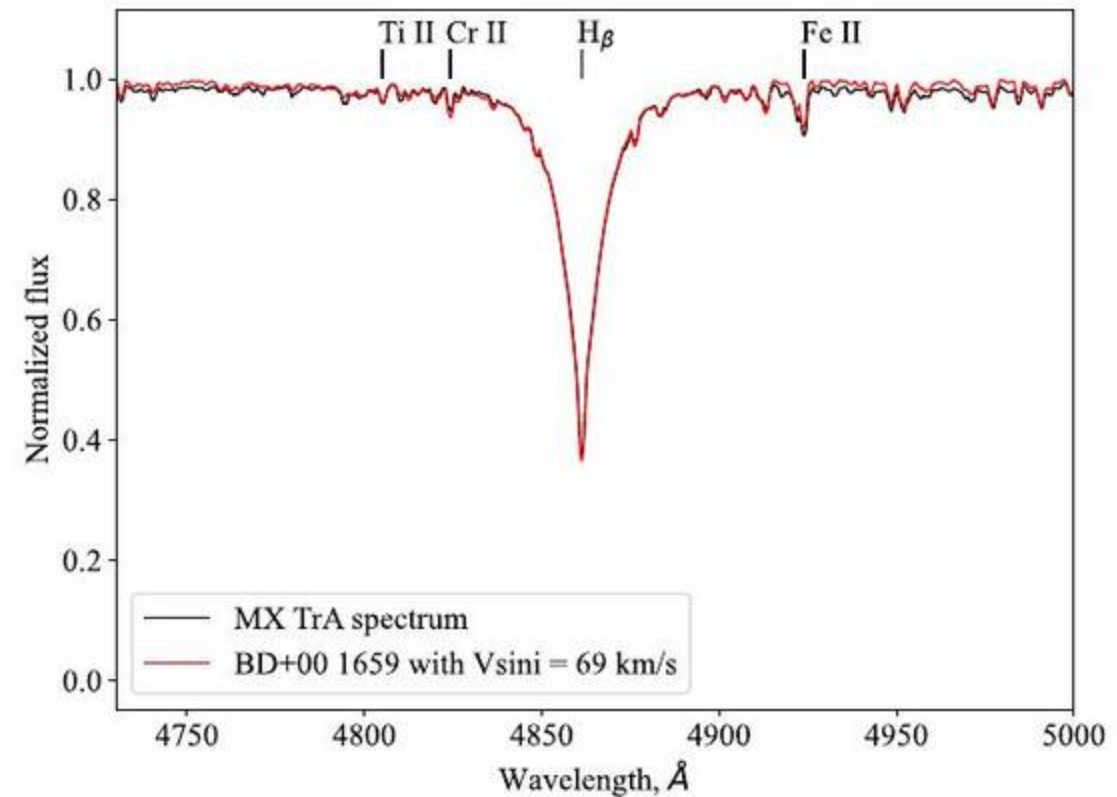
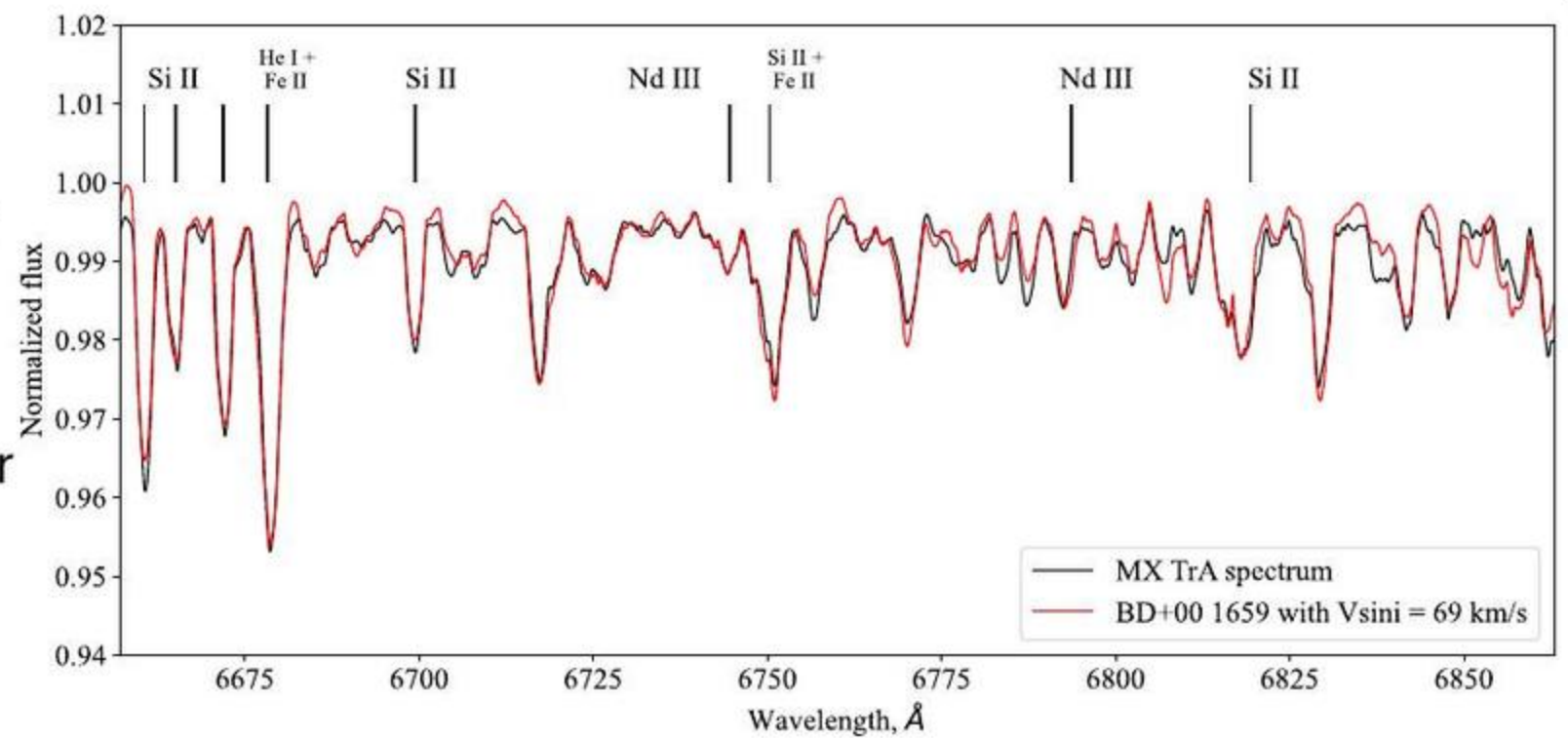
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**MX TrA****BD +00°1659**



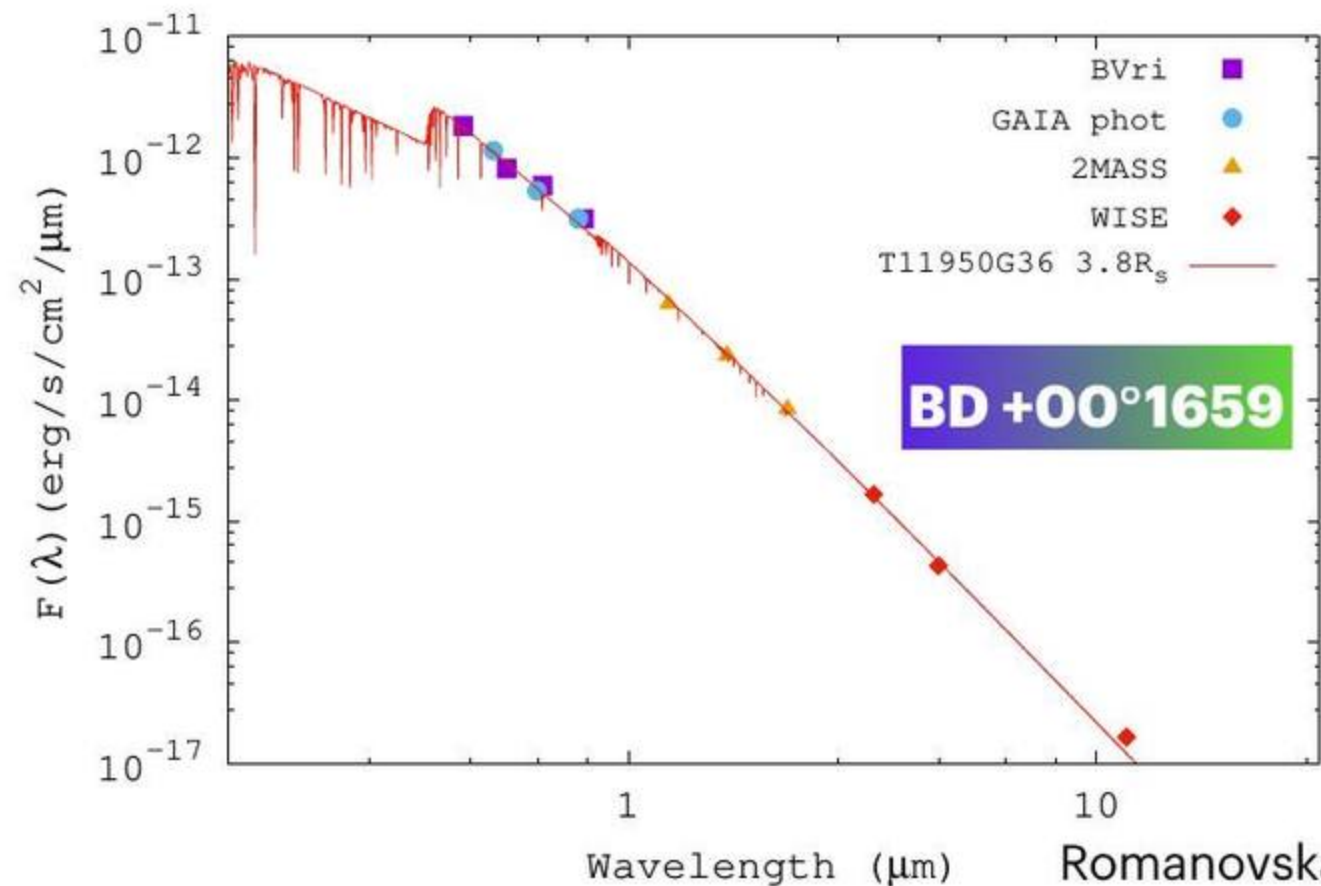
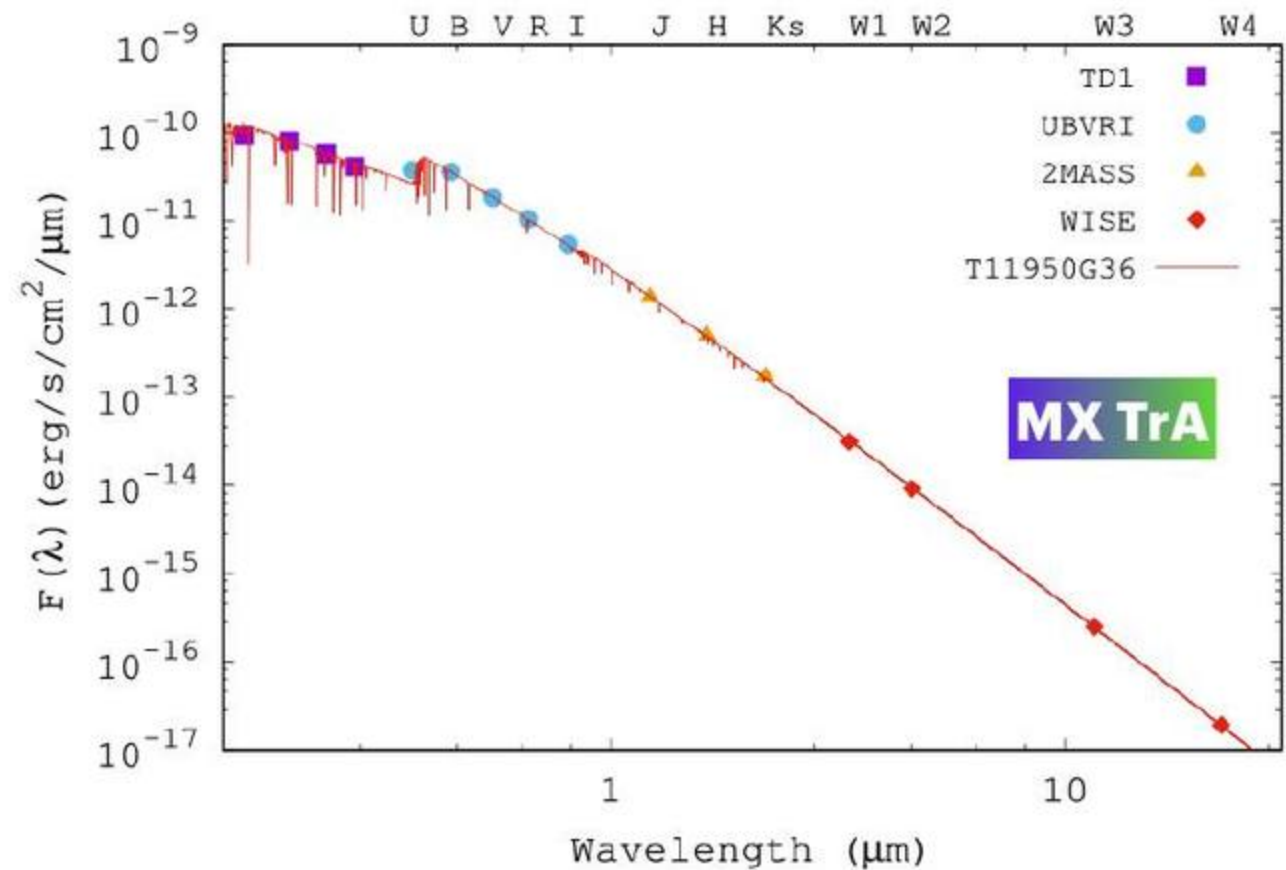
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- SED can be described by the same model atmosphere

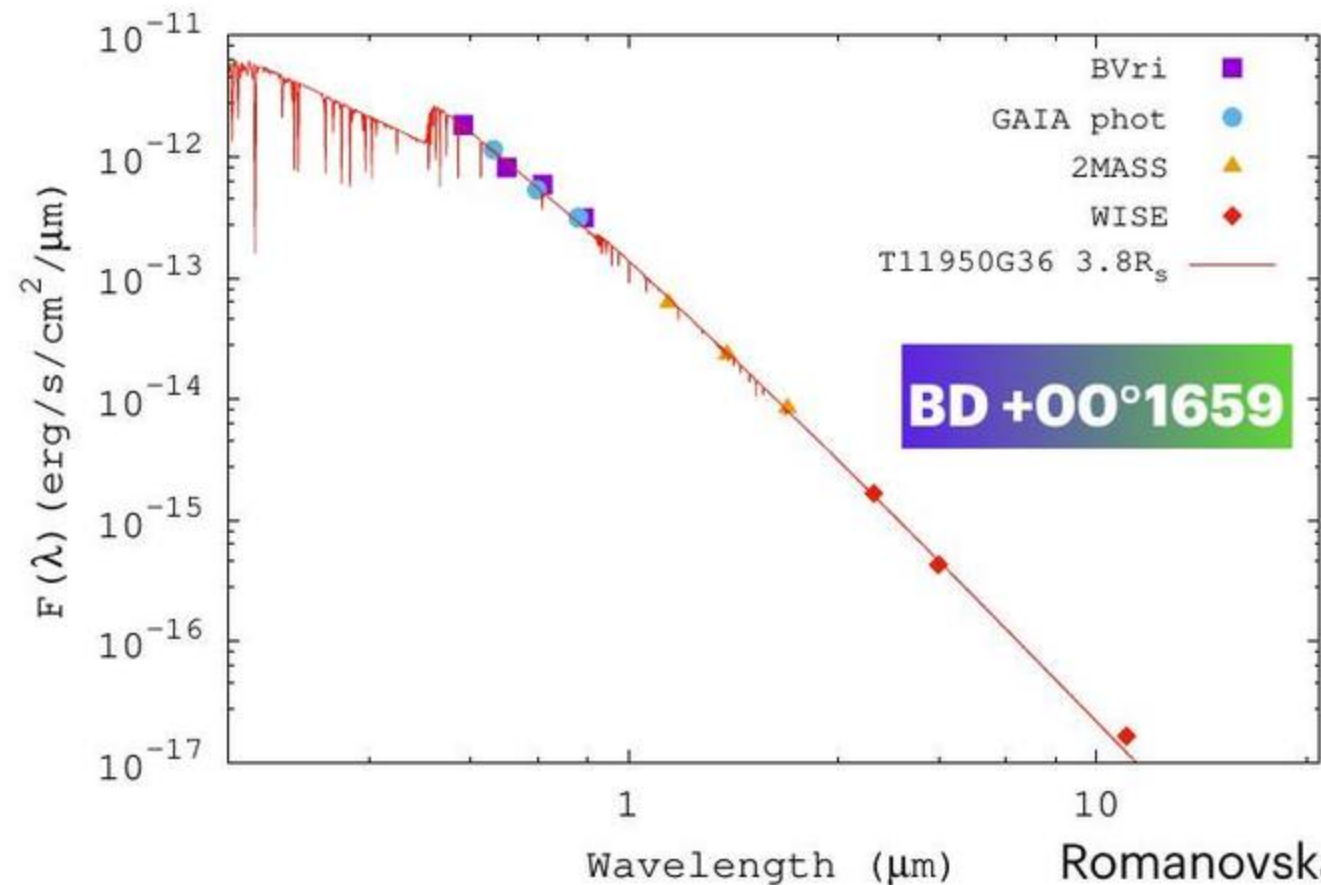
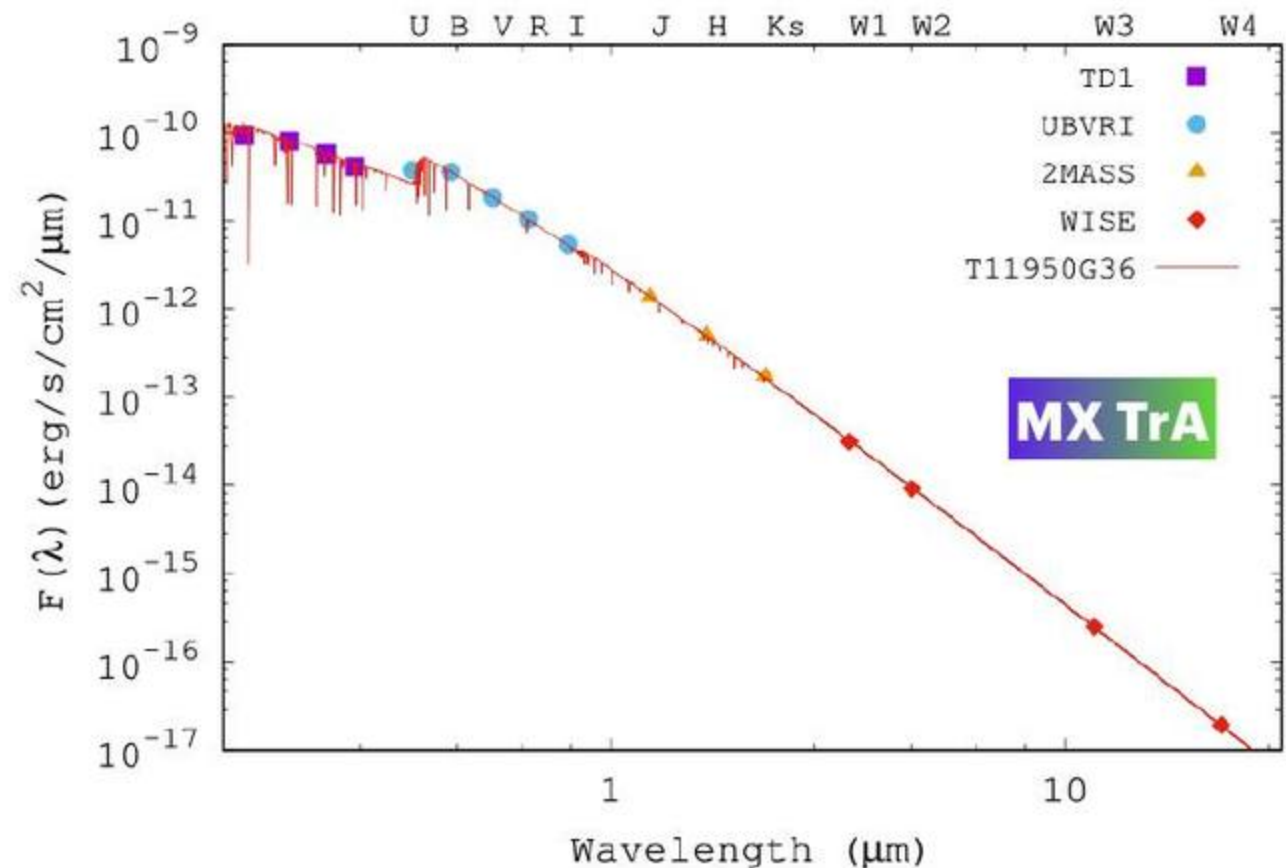




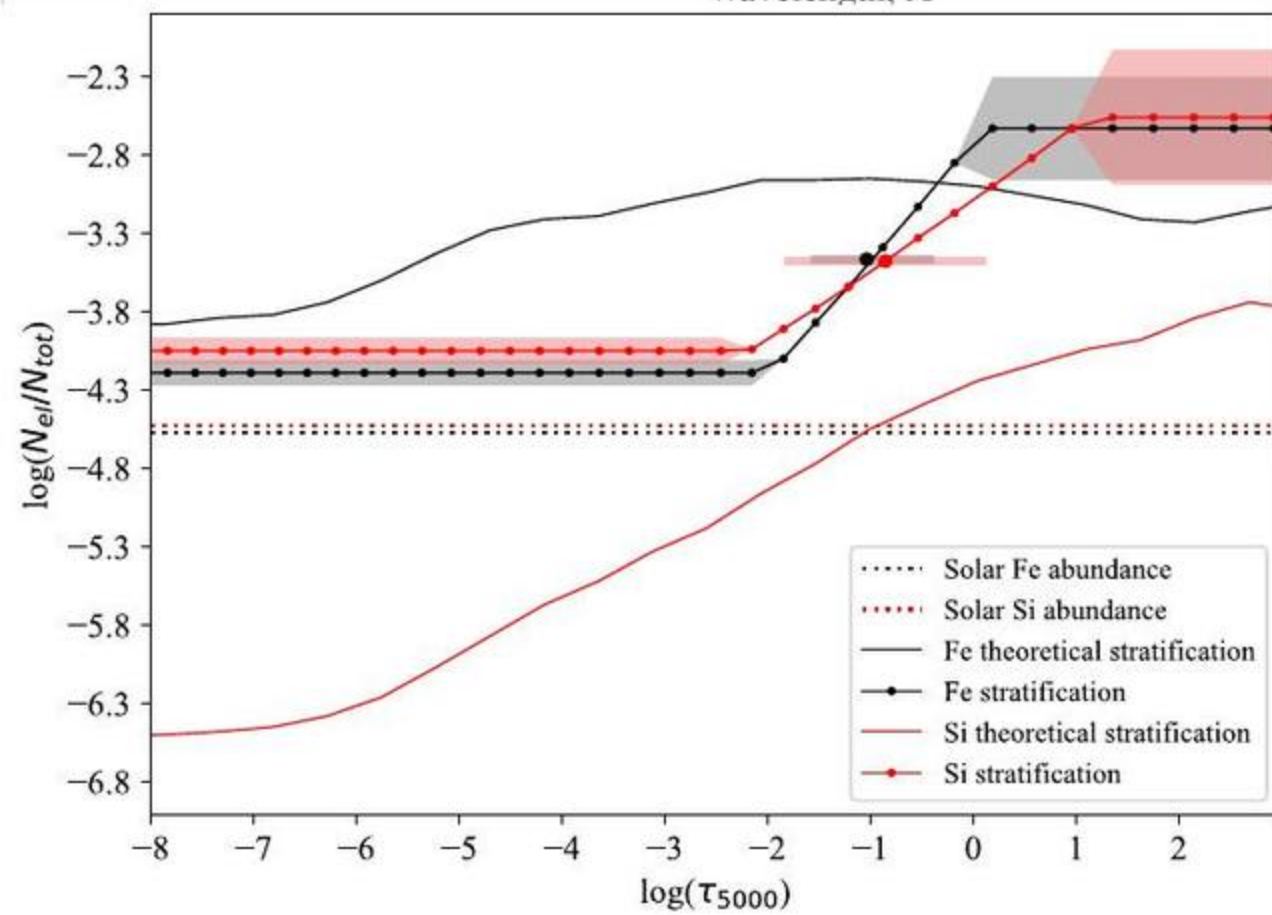
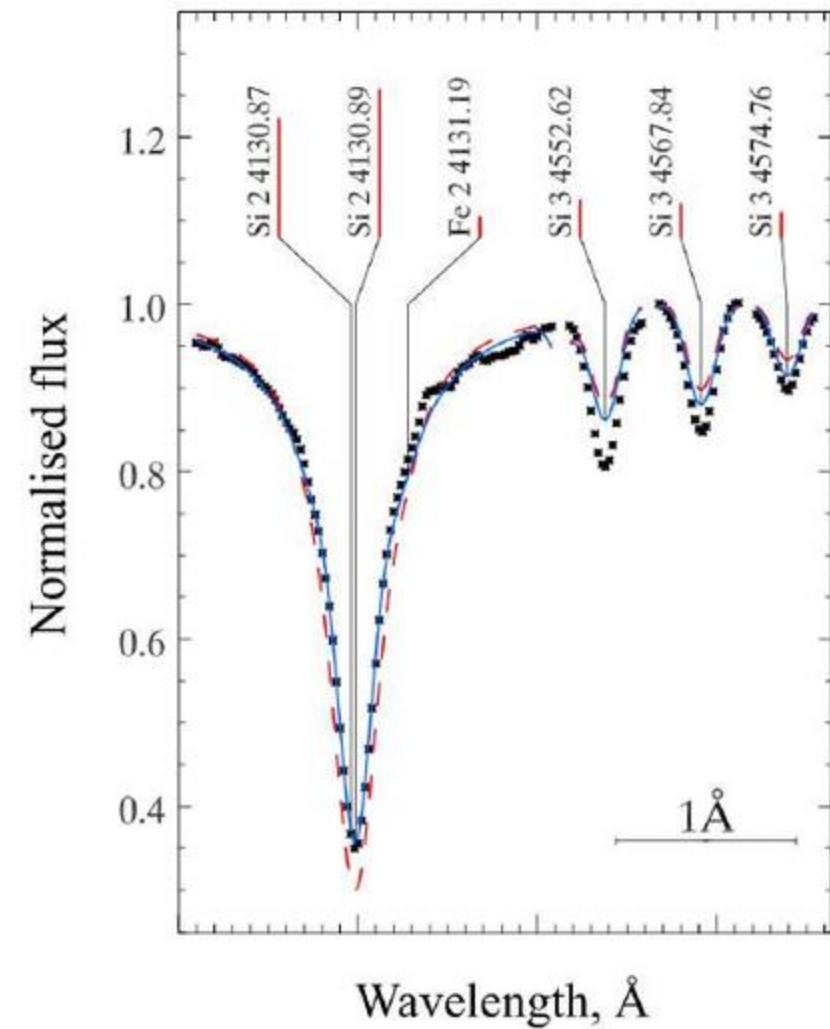
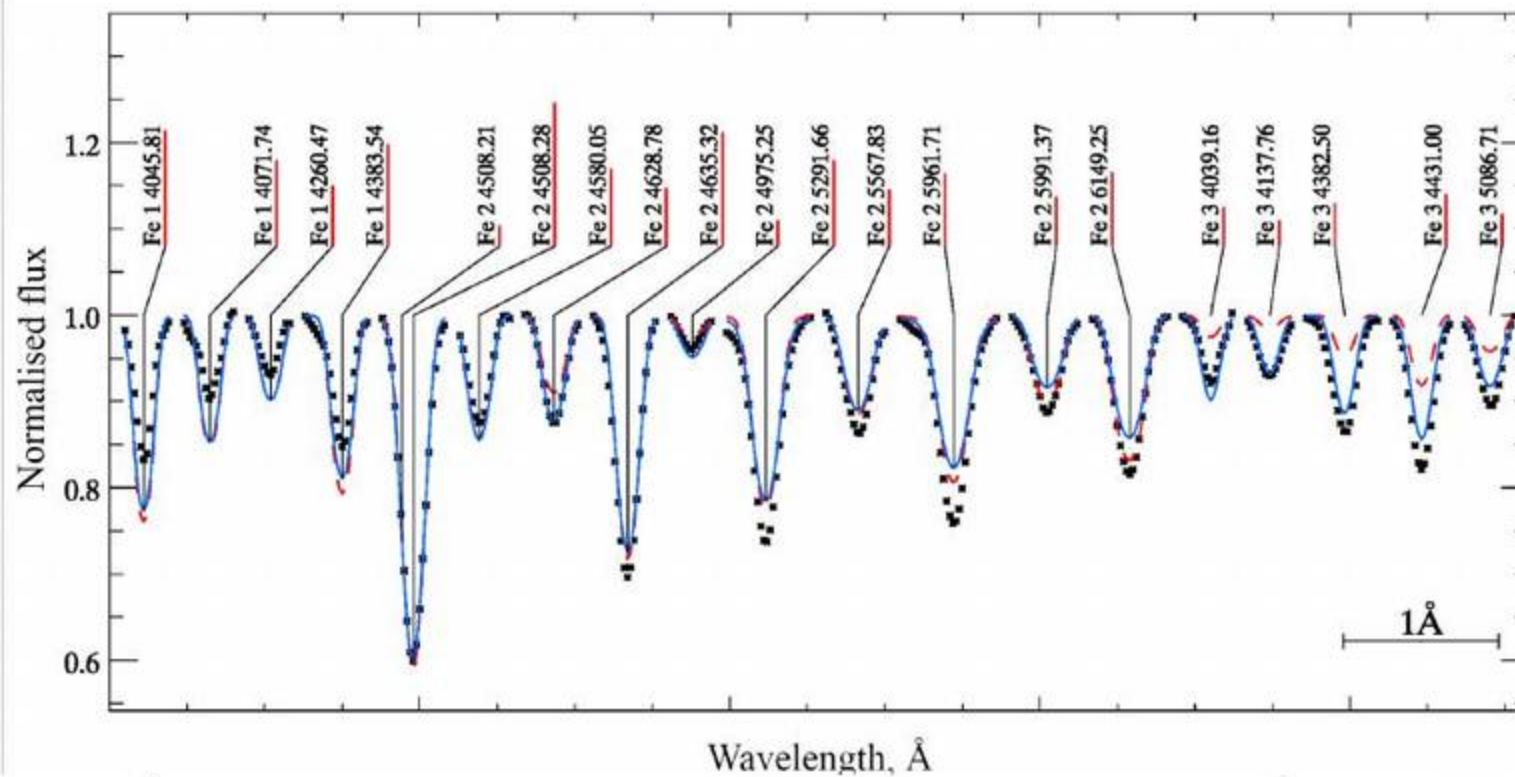
# BD 00°1659

- The observed spectrum of MX TrA and BD +00°1659 are similar, taking into account  $v \sin i = 69$  km/s for BD +00°1659
- SED can be described by the same model atmosphere

BD+00°1659 is the slowly rotating twin of MX TrA. Stratification profile in BD+001659 can be applied for MX TrA



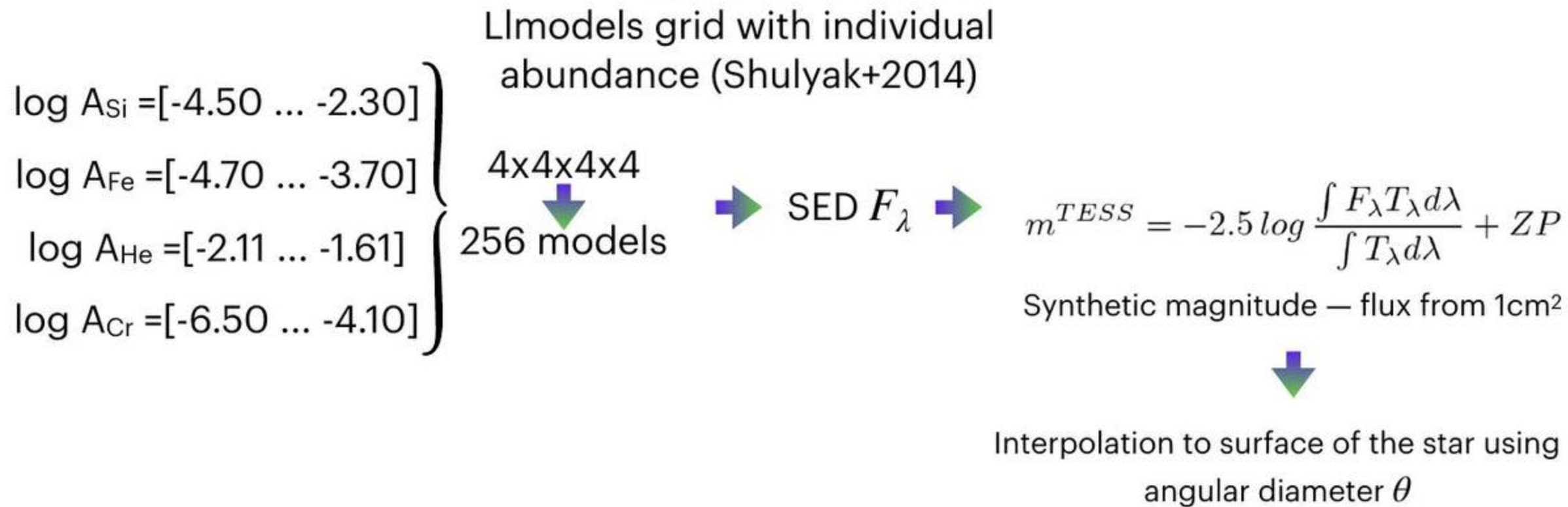
# Stratification of Fe and Si



Theoretical stratification  
- LeBlanc+2009



# Calculation of intensity map of MX TrA



# Calculation of intensity map of MX TrA

Llmodels grid with individual  
abundance (Shulyak+2014)

$$\log A_{\text{Si}} = [-4.50 \dots -2.30]$$

$$\log A_{\text{Fe}} = [-4.70 \dots -3.70]$$

$$\log A_{\text{He}} = [-2.11 \dots -1.61]$$

$$\log A_{\text{Cr}} = [-6.50 \dots -4.10]$$

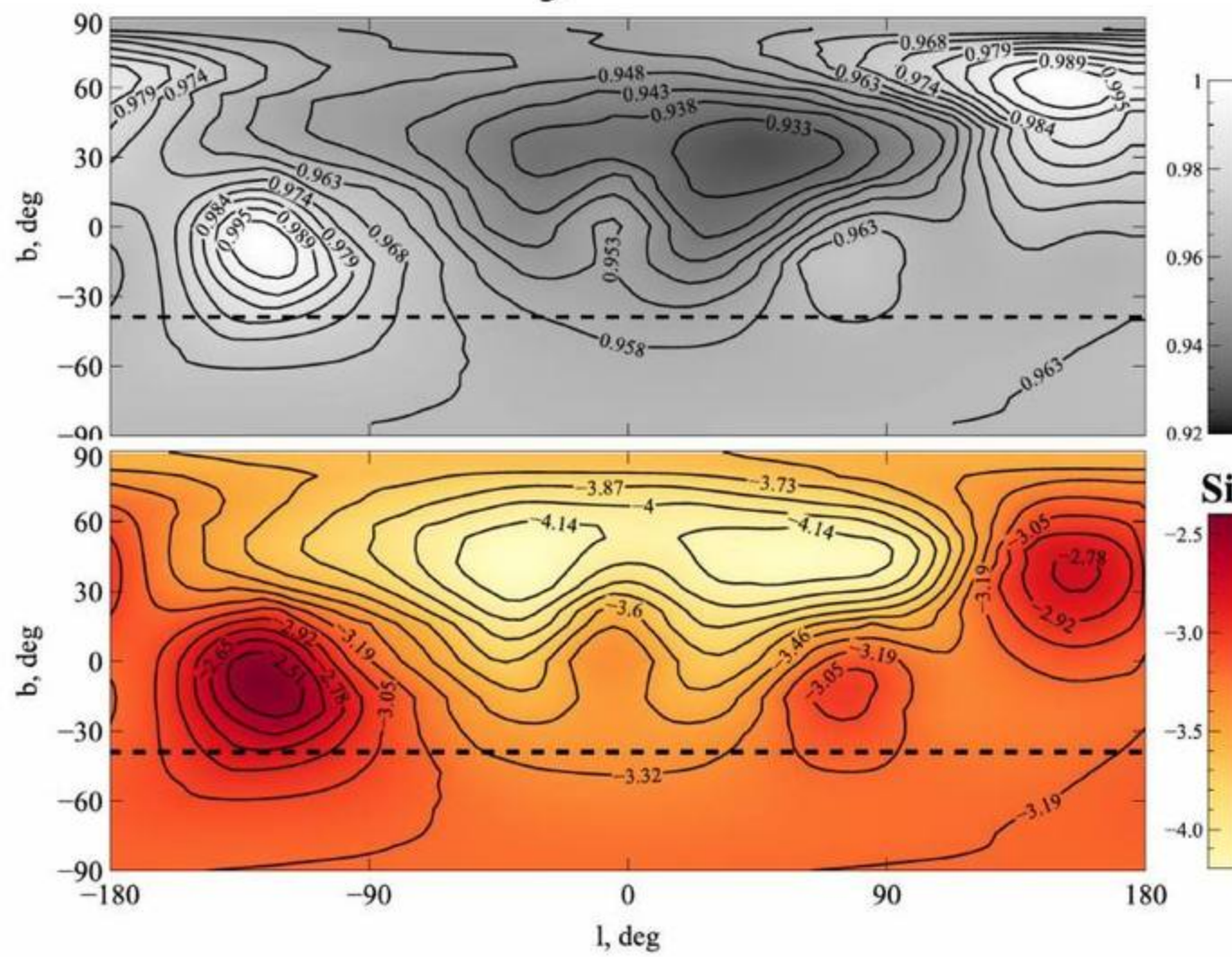
4x4x4x4

256 models

SED  $F_{\lambda}$

$$m^{TESS} = -2.5 \log \frac{\int F_{\lambda} T_{\lambda} d\lambda}{\int T_{\lambda} d\lambda} + ZP$$

Synthetic magnitude — flux from 1cm<sup>2</sup>



Interpolation to surface of the star using  
angular diameter  $\theta$

Derive intensity map



# Calculation of intensity map of MX TrA

$\log A_{\text{Si}} = [-4.50 \dots -2.30]$   
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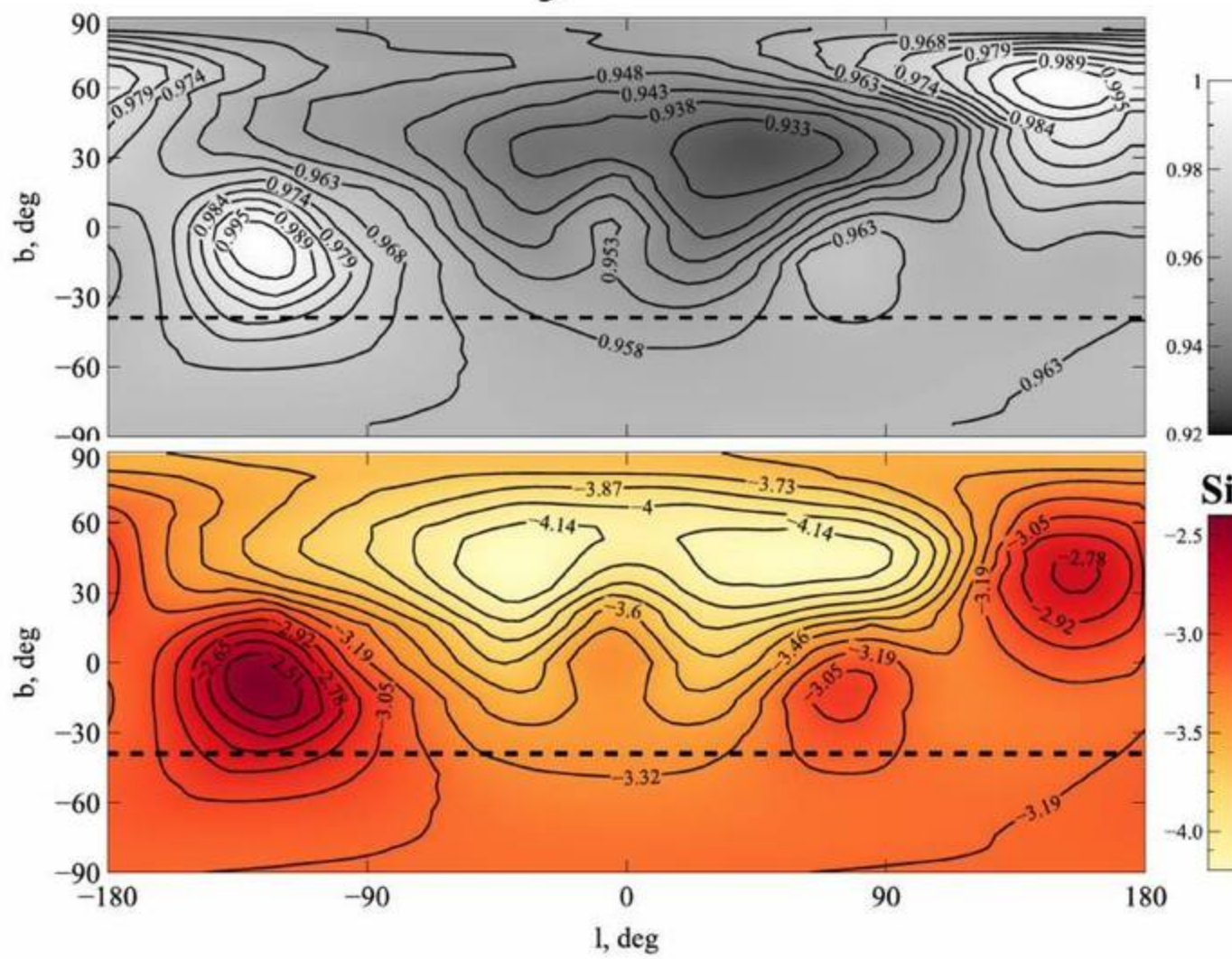
Lmodels grid with individual  
 abundance (Shulyak+2014)

$4 \times 4 \times 4 \times 4$   
 $\downarrow$   
 256 models

$\rightarrow$  SED  $F_{\lambda}$   $\rightarrow$

$$m^{TESS} = -2.5 \log \frac{\int F_{\lambda} T_{\lambda} d\lambda}{\int T_{\lambda} d\lambda} + ZP$$

Synthetic magnitude — flux from 1cm<sup>2</sup>

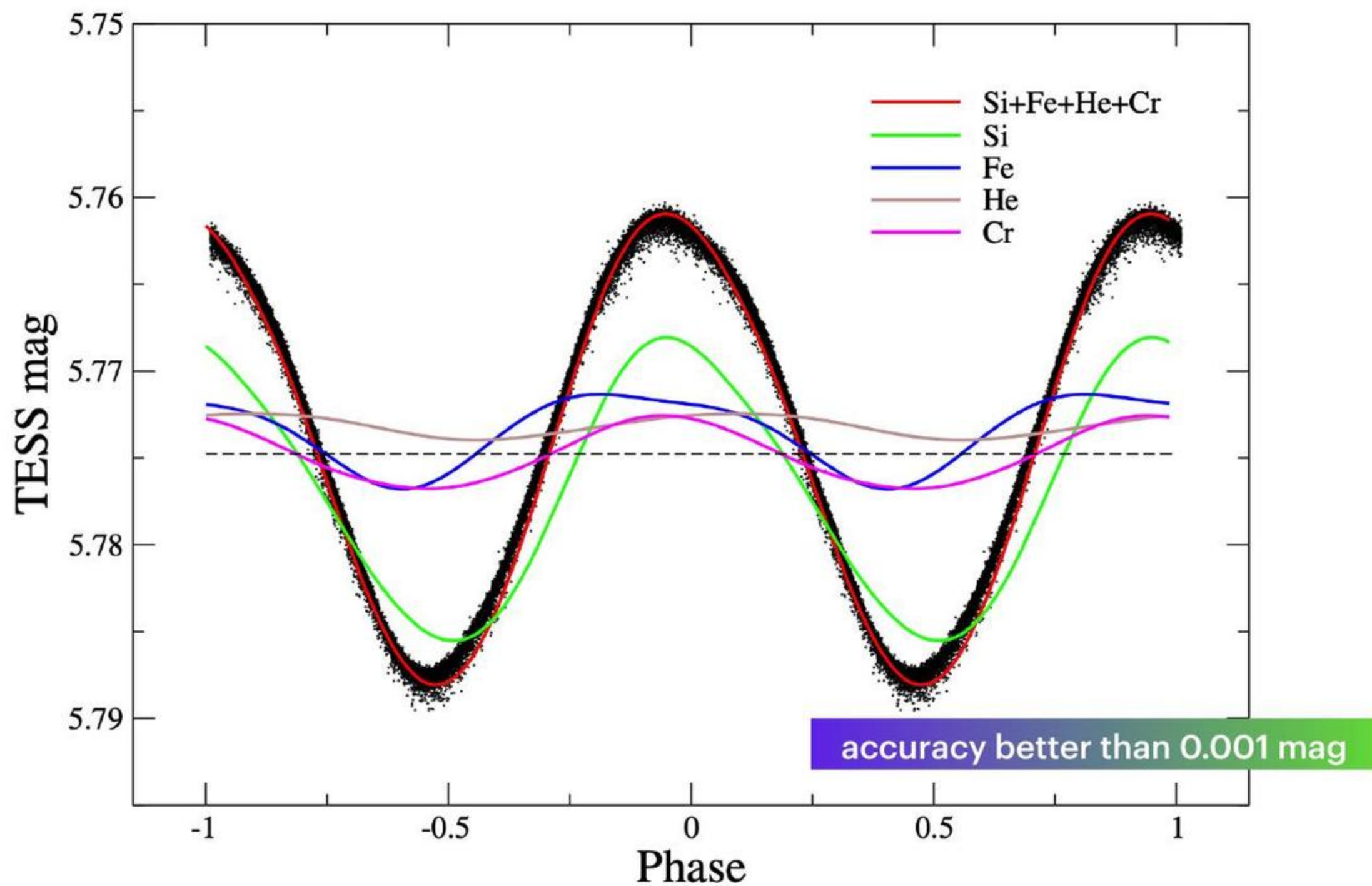


Interpolation to surface of the star using  
 angular diameter  $\theta$

Derive intensity map

The intensity map is most similar to  
 the silicon distribution map, since  
 Si makes the most significant  
 contribution to the absorption  
 coefficient

# Calculation of light curve of MX TrA





# Conclusion

1. Using high-resolution spectroscopy with SALT telescope we determined atmospheric parameters, abundances and performed Doppler Imaging of southern ApSi star MX TrA.
2. Abundance maps revealed highly inhomogeneous surface distribution of He, O, Mg, Si, Fe and Cr in MX TrA.
3. For accounting effects of vertical abundance stratification we employed its profile from slowly rotating twin star BD+00°1659.
4. A synthetic light curve was calculated taking into account 3D distribution of elements in MX TrA atmosphere. Theoretical light curve fits observational TESS data with accuracy better than 0.001 mag. Hence, inhomogeneous distribution of He, Si, Fe, Cr completely explains photometric variability of MX TrA.
5. Collaboration with BRICS countries has an outstanding perspectives for studying of southern sky with high-resolution spectroscopy.

**Thank you for  
your attention!**



## Results are published in papers:

1. I. Potravnov, T. Ryabchikova, N. Piskunov, Y. Pakhomov, A. Kniazev, “Doppler imaging of a southern ApSi star HD 152564”, MNRAS, vol.527, p.10376, 2024
2. Y. Pakhomov, A. Romanovskaya, I. Potravnov, T. Ryabchikova, “Modeling the TESS Light Curve of Ap Si Star MX TrA”, Universe, vol.10(9), p.341, 2024
3. A. Romanovskaya, T. Ryabchikova, Y. Pakhomov, I. Potravnov, T. Sitnova, “Slowly-rotating peculiar star BD+00°1659 as a benchmark for stratification studies in Ap/Bp stars”, Galaxies, submitted, 2024

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